Unit – V

Dr. P Arunkumar

Visual Analytics in Healthcare

Interpreting Complex Data for Better Outcomes

The Data Deluge

Healthcare generates massive, complex datasets from EHRs, genomics, imaging, and wearables. This data is often:

- **Voluminous:** Petabytes of information being generated and stored, far exceeding human capacity to analyze manually.
- **Varied:** A mix of structured (lab results, vitals) and unstructured (clinical notes, images) data that is difficult to unify.
- **Velocity:** Generated at high speed from real-time patient monitors and wearable devices, requiring immediate analysis.

Traditional analysis fails to capture the full picture.

What is Visual Analytics?

Automated Analysis

Using data mining and machine learning to pre-process data and automatically detect potential patterns.

Interactive Visualization

Presenting data in an intuitive, visual format that allows users to explore, filter, and drill down into details.

Human-in-the-Loop

Empowering clinicians with tools to apply their expertise, ask new questions, and guide the analytical process.

Technique: Trend Analysis

Visualizing admission rates helps in resource allocation and identifying seasonal peaks or outbreaks.

Application: Public Health

Tracking Outbreaks

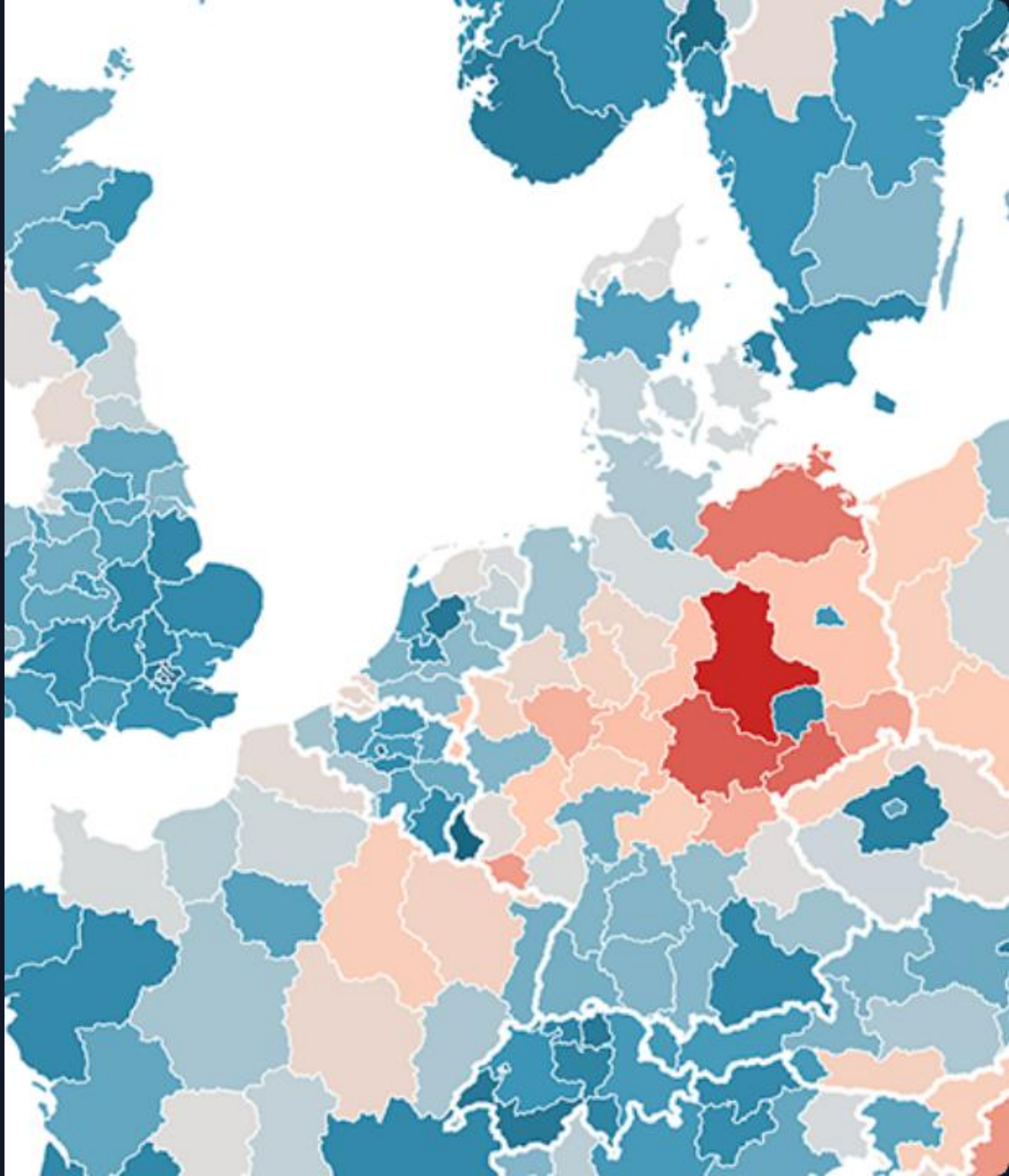
Interactive maps and network graphs visualize the spread of infectious diseases in real-time, helping public health officials respond quickly.

Identifying Hotspots

Geospatial heatmaps show geographic areas with high case concentration, guiding targeted interventions and resource deployment.

Vaccine Distribution

Dashboards monitor vaccination rates, inventory, and equitable access to ensure effective public health campaigns.



Application: Clinical Care

Personalized Medicine

Visualizing a patient's genomic data, lab results, and medical history on a single interactive timeline. This helps clinicians identify the best treatment path tailored to the individual.

Risk Stratification

Using visual models to analyze thousands of variables (vitals, labs, history) to flag patients at high risk for conditions like sepsis or readmission, enabling proactive intervention.

The Impact

Faster

Decisions

- **Improved Patient Outcomes:** Earlier detection of high-risk patients leads to better, more proactive care.
- **Operational Efficiency:** Optimizing hospital workflows, staff schedules, and resource management.
- **Democratized Data:** Making complex data accessible and understandable to clinicians, not just data scientists.